

Tarun Mundhra

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EDUCATION

Indian Institute of Technology, Kharagpur

Kharagpur, India

B.S.(Hons.) in Exploration Geophysics; CGPA: 8.13/10

Aug 2023 – May 2027

- **Relevant Coursework:** Linear Algebra, Probability & Statistics, Programming & Data Structures, Advanced Calculus, Pattern Recognition, Partial Differential Equations, Economics, Project Engineering and Management

TECHNICAL SKILLS

Languages: JavaScript, Python, C++, C, HTML, CSS, SQL

Frameworks, Libraries & Databases: Node.js, Express.js, React.js, Next.js, React Native (Expo), Tailwind CSS, NativeWind, TanStack Query, JWT, MongoDB, PostgreSQL, IndexedDB, MySQL

Tools & Technologies: Git, GitHub, RESTful APIs, Postman, AWS Lambda, Cloudinary, Multer, Service Workers

PROJECTS

KGP Presence – Academic Tracker Mobile App | *React Native, MongoDB* | *Lead Developer*

Jan 2026

- Architected a cross-platform attendance tracker using **React Native (Expo)** and **AWS Lambda**; drove the project from ideation to a fully functional preview build, currently slated for Google Play Console publication
- Co-developed an OCR-powered timetable generator extracting schedule data from uploaded images, alongside a customizable manual creation mode for full student flexibility
- Streamlined course integration via an **auto-fetch mechanism** querying the database by Subject Code to instantly populate details through an animated NativeWind UI
- Designed a one-click attendance logging system feeding into granular, real-time analytics filtered by semester, subject, and timetable, optimized with **TanStack Query**
- Built a dynamic user dashboard visualizing attendance metrics, integrated with an event-tracking notification system, enabling voice controls via **Sarvam STT** and **Groq API** JSON transcript parsing
- Secured the Node.js/Express MVC backend using a serverless MongoDB architecture, implementing **JWT**, **bcrypt**, and **OTP-based email verification** for robust authentication

Algorithmic Trading Simulator | *Go, Redis, PostgreSQL, React, WebSockets* | *Team Project*

Apr, 2026

- Built automated market agents including a synthetic market maker for baseline liquidity and a noise trader simulating retail order flow with momentum signals with single click **Docker containerisation**
- Developed a deterministic limit order book and matching engine enforcing strict **price-time priority**, with Self-Trade Prevention (STP) to detect and cancel wash-trading orders by User ID
- Implemented a real-time data pipeline using **Go channels** and **Redis Pub/Sub** to stream Level-2 order book snapshots, live candlestick updates, and portfolio recalculations via WebSockets to a React frontend
- Ensured reliable persistence by storing transactional data in PostgreSQL and partitioning high-frequency trade records in TimescaleDB for efficient historical querying

TaskFlow Pro | *Node.js, Express.js, Graph Algorithms, Priority Queue, JWT*

Dec 2025

- Developed a full-stack task management platform with graph-based **dependency management**, modeling tasks as nodes and dependencies as directed edges with dynamic graph reconstruction
- Implemented cycle detection and **topological sorting** ($O(V + E)$ complexity) to compute valid task order
- Architected **Priority Queue**-driven scheduling engine using heap data structures to dynamically rank tasks based on priority, deadlines, and dependency readiness
- Optimized backend performance by validating affected subgraphs, ensuring consistency across modifications

POSITIONS OF RESPONSIBILITY

Vice-Captain – Interhall GC-OpenSoft Competition

MMM Hall Of Residence, IIT Kharagpur

Hall Technical Committee

2026

- Led a 20-member cross-functional development team to design and deliver a full-stack, institutional-grade algorithmic trading simulator under competitive inter-hall conditions

ACHIEVEMENTS

Secured **Top 1%** among 1.2M candidates in JEE Main 2023 and **Top 6%** among 200K in JEE Advanced 2023

Completed NCC 'B' Certificate and certified training camp involving drills and leadership development